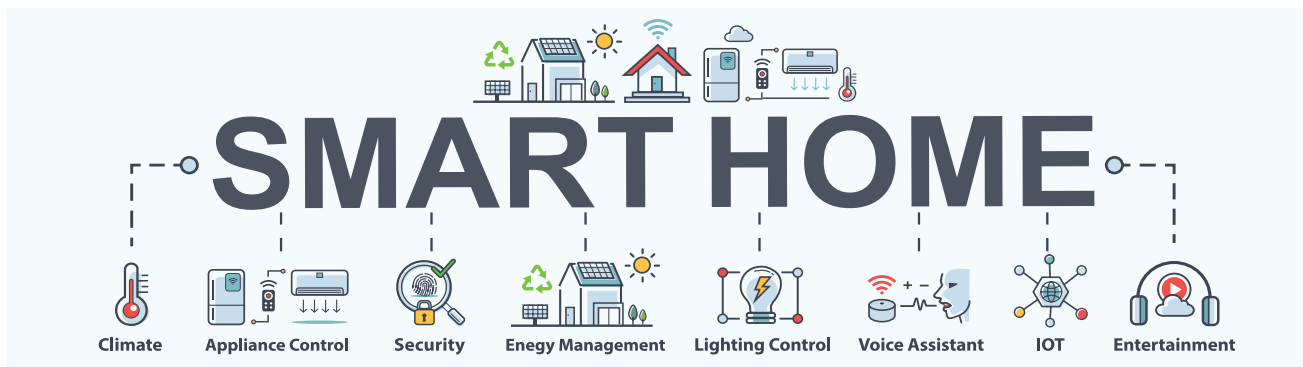




and the device is available in other apps like Google Home as well. For example, you can check your doorbell with Alexa, Google Home, etc at the same time with minimal configuration. This is the kind of flexibility that only Matter offers at present.

Protocol stack

Matter is an application layer protocol that is very flexible, as it does not require you to change your Wi-Fi card or a ZigBee card. Matter can support Wi-Fi and Thread devices.

Most of the code that is needed is available in open source repositories and you do not have to invest in software development. You can either get code from the repositories themselves or you can go to chip vendors, who can give you the chip with Matter binary loaded in it.

Certification process

The certification of Matter involves following steps:

1. Become a member. Read Connectivity Standards Alliance Policies and Governing Documents.

2. Request a manufacturer/vendor ID. Contact the Alliance Certification Team to reserve your manufacturer ID or vendor ID.

3. Select a Compliant Platform or Network Transport. For Zigbee and Smart Energy end products, there is access to Compliant Platforms from multiple silicon vendors. Select one from the list of certified Compliant Platforms. For Matter end products, you have the

choice of various network transport options, including Wi-Fi, Thread, and Ethernet. Note that you will need to obtain certification for your network transport from the relevant standards organisation before you certify your Matter device with Alliance.

4. Choose a Testing Provider.

Select from Connectivity Standards Alliance authorised Test Providers at locations all around the world.

5. Send product to be tested. After scheduling testing with an authorised Test Provider, the facility will make arrangements for testing samples and Protocol Implementation Conformance Statement (PICS) documents to be submitted. Test Provider will issue a final report to the Connectivity Standards Alliance when testing is successfully completed.

6. Submit Certification Application. Complete and apply for certification in Connectivity Standards Alliance Certification Tool. Instructions for requesting a Certification Tool account and creating/submitting applications can be found in the Connectivity Standards Alliance Members Area.

7. Application Pending. Connectivity Standards Alliance Certification Team will review your application and, if necessary, request action on specific identified items or information required to decide approval or rejection. At any time during this process, you may reach out to the Connectivity Standards Alliance Certification Team with any questions.

8. Approval. After your product certification is approved, you will receive a formal certificate from Alliance and you may now start using the Certified Product logo. Certified Product logos have usage guidelines that govern how they are used. Please review the applicable sections before affixing. Logos are provided to the applicant's contact for the certification. Matter products will also receive a Certification Declaration (CD) blob file and a certified product record entered in the Distributed Compliance Ledger (DCL).

To conclude, Matter has redefined the trajectory of smart home technology, offering a glimpse into a future where compatibility and connectivity seamlessly coexist. With its ability to simplify device integration and promote interoperability, Matter opens up a world of possibilities for creating truly smart and interconnected homes.

As we continue to witness advancements in this field, embracing the Matter standard will be instrumental in unlocking the full potential of smart homes, making our lives more convenient, efficient, and enjoyable than ever before. So, get ready to rediscover the future of the smart home ecosystem with Matter at the helm. **EFY**

This article is based on the tech talk by Ravindra B. Shet, Senior Director, Head of IoT Team, Samsung R&D Institute India Bangalore (SRIB) Pvt Ltd. It has been transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY, who has a knack for open source exploration and research